

A COURSE IN LINEAR ALGEBRA

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Algebra Lectures

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CONTENTS

1	VECTOR SPACES	1
1.1	Linear Combinations and Spanning Sets	2
1.1.1	Characterization of Basis	3
1.2	Basis and Dimension	4
1.3	Linear Transformations	6
1.3.1	Matrix Representation	7
1.3.2	Isomorphisms of Vector Spaces	8
1.3.3	Operators and Projections	10
2	MATRICES	13
2.1	Construction of The Determinant Tensor	13
2.2	Geometric Interpretation of the Determinant Map	16
2.3	Multiplicativity of the Determinant	17
2.4	The Adjoint Matrix	19
2.4.1	Matrix Reprsentations of Homomorphisms	22
3	SYSTEMS OF LINEAR EQUATIONS	23
3.1	The Dual Vector Space and Dimension	23
3.2	Row and Columns Spaces	24

VECTOR SPACES

In this book we present a unified theory of linear algebra, in which we present the topics of vector spaces, linear transformations and matrices in a natural fashion. We will begin with the notion of a vector space over a general field $\mathbb{K}$. It is assumed that the reader has some familiarity with high-school algebra and the real numbers $\mathbb{R}$, preferably the complex numbers $\mathbb{C}$, although this is certainly not required. Some knowledge of group theory is helpful, but is not crucial; groups only help one to visualize vector spaces. This book is essentially self-contained, and the required theorems for groups will be given here as well. Recall the definition of a group:

Definition 1. A group is a pair $(G, \odot)$ where G is a set and $\odot : G \times G \rightarrow G$ is a function satisfying

1. There exists $e \in G$ so that $e \odot g = g \odot e = g$ for all $g \in G$.
2. For all $g \in G$ there is some $g^{-1} \in G$ so that $g \odot g^{-1} = g^{-1} \odot g = e$

We often write $gf = g \odot f$ for $f, g \in G$ to ease notation.

A group G is said to be **Abelian** if it is commutative, i.e. $gh = hg$ for all $g, h \in G$. Many linear algebra courses study vector spaces over $\mathbb{R}$, or sometimes $\mathbb{C}$ (the concept of *over* will be defined soon). There is no real (heh) reason to do so, other than familiarity with $\mathbb{R}$. When the subject is developed over $\mathbb{C}$ rather than $\mathbb{R}$, we gain some generality since $\mathbb{R} \subset \mathbb{C}$. In reality, to develop this theory we need only restrict ourselves to a field $\mathbb{K}$. Now, recall the definition for a field $\mathbb{K}$:

Definition 2 (Fields). A field, oft denoted $\mathbb{K}$, is a set equipped with two binary operations $+, \cdot : \mathbb{K} \times \mathbb{K} \rightarrow \mathbb{K}$ satisfying for all $a, b, c \in \mathbb{K}$:

- (a) $(a + b) + c = a + (b + c)$.
- (b) $a + b = b + a$.
- (c) There exists an element $0 \in \mathbb{K}$ such that $a + 0 = 0 + a = a$.
- (d) There exists an element $(-a) \in \mathbb{K}$ such that $a + (-a) = (-a) + a = 0$. We shall often write $-a$ and $a - a = a + (-a)$ in this case.
- (e) $a(bc) = (ab)c$.
- (f) $(a + b)c = ac + bc$.

(g) There exists an element $1 \in \mathbb{K}$ such that $1(a) = a(1) = a$.

(h) If $a \neq 0$ then there exists $a^{-1} \in \mathbb{K}$ such that $a(a^{-1}) = a^{-1}(a) = 1$. The non-zero elements of $\mathbb{K}$ are called the **units** of $\mathbb{K}$ and are denoted by $\mathbb{K}^\times$.

Now we may define a vector space V over a field $\mathbb{K}$:

Definition 3. A vector space V is a set endowed with two operations

$$\begin{aligned} + : V \times V &\rightarrow V, & (u, v) &\mapsto u + v \\ \odot : V \times \mathbb{K} &\rightarrow V, & (u, \alpha) &\mapsto \alpha u \end{aligned}$$

such that $(V, +)$ and $(V, \odot)$ are abelian groups.

We refer to the elements of V as *vectors*. The two operations above are called vector addition and scalar multiplication, respectively. The simplest example is $\mathbb{R}$, with usual addition and multiplication. The most frequent example is d -dimensional Euclidean space: $\mathbb{R}^d$. A set $W \subseteq V$ is said to be a **subspace** of V provided W is closed under (vector) addition and closed under scalar multiplication. That is, for any $u, v \in W$ and $\alpha \in \mathbb{K}$ one has $u + v \in W$ and $\alpha u \in W$.

1.1 LINEAR COMBINATIONS AND SPANNING SETS

Throughout this section we assume we have a vector space V over a non-empty field $\mathbb{K}$. Consider now a non-empty set $\mathcal{S} \subseteq V$. What can be done with the elements in $\mathcal{S}$? Clearly, since V is a vector space we may add and “scale” these elements in $\mathcal{S}$, and this leads us to the following definition:

Definition 4. A **linear combination** of elements in $\mathcal{S} \subseteq V$ is a finite sum over $\mathbb{K}$: $\sum_{j=1}^n \alpha_j u_j$ where $u_j \in \mathcal{S}$ and $\alpha_j \in \mathbb{K}$ for $1 \leq j \leq n$. The **span** of $\mathcal{S}$, denoted $\text{span}(\mathcal{S})$, is the collection of all linear combinations in $\mathcal{S}$:

$$\text{span}(\mathcal{S}) = \left\{ \sum_{j=1}^n \alpha_j u_j : u_j \in \mathcal{S}, \alpha_j \in \mathbb{K} \text{ and } n \in \mathbb{N} \right\} \quad (1)$$

Our first result is that a set $\mathcal{S}$ induces a subspace of V in its span:

Proposition 1.1. Let $\mathcal{S} \subseteq V$ be a non-empty set. Then, $\text{span}(\mathcal{S})$ is a subspace of V .

Proof. Let $u, v \in \text{span}(\mathcal{S})$. Then, by letting some coefficients be 0 as required we may assume that u, v are combinations of the very same vectors in $\mathcal{S}$, i.e. $u = \sum_{j=1}^n \alpha_j u_j$ and $v = \sum_{j=1}^n \beta_j u_j$ for $u_j \in \mathcal{S}$ and $\alpha_j, \beta_j \in \mathbb{K}$. Now, we find that $u + v = \sum_{j=1}^n (\alpha_j + \beta_j) u_j = \sum_{j=1}^n \gamma_j u_j \in \text{span}(\mathcal{S})$ by letting $\gamma_j = \alpha_j + \beta_j \in \mathbb{K}$.

Now to show closure under scalar multiplication we employ a similar argument. For $u \in \text{span}(\mathcal{S})$ we write $u = \sum_{j=1}^n \alpha_j u_j$ and observe that $\varepsilon u = \sum_{j=1}^n (\varepsilon \alpha_j) u_j$ which clearly lives in $\text{span}(\mathcal{S})$; where $\varepsilon \in \mathbb{K}$. Therefore $\text{span}(\mathcal{S})$ is a subspace of V as was asserted.

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We say a set $\mathcal{S}$ **spans** a space V whenever $\text{span}(\mathcal{S}) = V$. We call this same set **minimal spanning** if $\mathcal{S} \setminus \{u\}$ no-longer spans V for any $u \in \mathcal{S}$. This set $\mathcal{S}$ is said to be **linearly independent** provided the equality $\sum_{j=1}^n \alpha_j u_j = 0$ implies that $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$ for all finite subsets $\{u_j\}_j \subseteq \mathcal{S}$. This set is called **linearly dependent** if it is not independent. $\mathcal{S}$ is **maximally independent** if and only if $\mathcal{S} \cup \{u\}$ is linearly dependent.

Lemma 1.2. *Let $\mathcal{S} \subseteq V$. Then $u \in \text{span}(\mathcal{S})$ if and only if $\text{span}(\mathcal{S}) = \text{span}(\mathcal{S} \cup \{u\})$.*

Proof. Fix $u \in \text{span}(\mathcal{S})$, clearly $\text{span}(\mathcal{S}) \subseteq \text{span}(\mathcal{S} \cup \{u\})$. To see the reverse inclusion, let $v \in \text{span}(\mathcal{S} \cup \{u\})$. Then, $v = \sum_{j=1}^n \alpha_j u_j + \beta u$, where $u_j, u \in \mathcal{S}$ and therefore $v \in \text{span}(\mathcal{S})$ and we have equality. Conversely, assume $\text{span}(\mathcal{S}) = \text{span}(\mathcal{S} \cup \{u\})$. Then since $u \in \text{span}(\mathcal{S} \cup \{u\})$ we have $u \in \text{span}(\mathcal{S})$.

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1.1.1 Characterization of Basis

We now introduce the notion of a basis for a vector space. Given a vector space V over a field $\mathbb{K}$ we say a subset $\mathcal{B} \subseteq V$ is a **basis** for V if $\mathcal{B}$ is a spanning set for V and $\mathcal{B}$ is linearly independent. An example is the set $\mathcal{B} := \{(1, 0), (0, 1)\}$ for $\mathbb{R}^2$.

We offer the following equivalence for a set $\mathcal{S} \subseteq V$

Theorem 1.3. *Let V be a vector space over $\mathbb{K}$. The following are equivalent:*

1. $\mathcal{S}$ is minimal spanning.
2. $\mathcal{S}$ is maximally independent.
3. Every vector $u \in V$ has a unique representation $\sum_{j=1}^n \alpha_j u_j$ as linear combinations of $\mathcal{S}$ where the α_j are non-zero.

Proof. (1 $\implies$ 2). Let $\mathcal{S}$ be minimally spanning. We first show that it is independent. Indeed, if not then we may find $\{u_1, u_2, \dots, u_n\} \subseteq \mathcal{S}$ so that $\sum_{j=1}^n \alpha_j u_j = 0$ but some $\alpha_j \neq 0$. After a reordering, we may assume $\alpha_n \neq 0$ and then recover $u_n = \alpha_n^{-1} \sum_{j=1}^{n-1} \alpha_j u_j = \sum_{j=1}^{n-1} \beta_j u_j$. This contradicts the fact that $\mathcal{S}$ is minimally spanning, for then we could “delete” u_n for $\mathcal{S}$ and retain $\text{span}(\mathcal{S}) = V$ since by Lemma 1.2 we find $\text{span}(\mathcal{S} \setminus \{u_n\}) = \text{span}(\mathcal{S}) = V$. We now show maximal independence. Let $v \in V$, if $v \in \mathcal{S}$ or $v = 0$ then we are done. Else, since $\mathcal{S}$ is spanning we may write $v = \sum_{j=1}^n \alpha_j u_j$ for $u_j \in \mathcal{S}$. Now, since $\sum_{j=1}^n \alpha_j u_j - v = 0$ is a non-trivial linear combination we are done.

(2 $\implies$ 3). Assume $\mathcal{S}$ is maximally independent and fix $u \in V$. Now, $\mathcal{S} \cup \{u\}$ is linearly independent and so $\sum_{j=1}^n \alpha_j u_j + \beta u = 0$ where at least one coefficient is non-zero. A moments reflection tells us that $\beta \neq 0$ for otherwise one would have $\alpha_j = 0$ for all j since $\mathcal{S}$ is independent. Therefore, we may isolate u by multiplying through with β^{-1} and neglecting 0 coefficients gives us a representation. To see uniqueness, write

$$u = \sum_{j=1}^n \alpha_j u_j = \sum_{j=1}^n \beta_j u_j$$

where we allow 0 coefficients. Now, $\sum_{j=1}^n (\alpha_j - \beta_j) u_j = 0$ and by independence of $\mathcal{S}$ we get $\alpha_j = \beta_j$ for all j which gives us uniqueness.

Finally, (3 $\implies$ 1). We are given that $\mathcal{S}$ is a spanning set for V . We must show that $\mathcal{S} \setminus \{u\}$ is no-longer spanning for $u \in \mathcal{S}$. This is easy, clearly u is the only representation of u as linear combinations of $\mathcal{S}$ by hypothesis. If $\text{span}(\mathcal{S} \setminus \{u\}) = V$ then there is a different combination of elements in $\mathcal{S}$ giving u , a contradiction.

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Corollary 1.4. *Let V be a vector space over $\mathbb{K}$. A subset $\mathcal{B}$ is a basis for V if and only if it satisfies all of the above.*

Proof. Let $\mathcal{B}$ be a basis. Clearly, it is spanning and linearly independent by definition. We claim that it is minimally spanning. If we pull $u \in \text{span}(\mathcal{B})$ then $u \notin \text{span}(\mathcal{B} \setminus \{u\})$ for otherwise u is a linear combination of elements in $\mathcal{B}$ other than itself and $\mathcal{B}$ is not independent. Conversely, suppose that $\mathcal{B}$ is minimally spanning. By Theorem 1.3 it is also independent.

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1.2 BASIS AND DIMENSION

Let again V be a vector space over a field $\mathbb{K}$. It is a non-trivial consequence of the axiom of choice that there exists a basis $\mathcal{B}$ for V . We shall omit this proof and instead study consequences. Now consider a vector space V over a field $\mathbb{K}$, we define the **dimension** of the vector space, denoted $\dim(V)$, to be $\#\mathcal{B}$ for a basis $\mathcal{B}$ of V . Here $\#(\cdot)$ denotes the cardinality of a set.

There are several things to do with this. First, we must prove that this is *well defined*. Namely, we need to show that any two basis of a vector space V have the same cardinality.

We begin with a useful lemma:

Lemma 1.5. *A family of vectors is linearly dependent if and only if some vector is a linear combination of those preceding it.*

The proof is easy and is left as an exercise to the reader (try to prove this!).

Let us now give a powerful lemma

Lemma 1.6 (Steinitz'). *If $\{a_1, a_2, \dots, a_n\}$ and $\{b_1, b_2, \dots, b_m\}$ are linearly independent with $n \leq m$ then for all $1 \leq j \leq n$ we may re-order the elements in $\{b_j\}_{j=1}^m$ so that the family*

$$\{a_1, a_2, \dots, a_j, b_{j+1}, \dots, b_m\}$$

is linearly independent.

Proof. We argue by induction on j . The case for $j = 0$ is obvious, as this is just our original set $\{b_j\}$, now assume that it holds up to j , if $j = n$ then we are done. Else, $j < n$ and we show that this holds for $j + 1$. Now, by our inductive hypothesis we have re-ordered the list so that $\{v_1, \dots, v_j, w_{j+1}, \dots, w_j\}$ is linearly independent. Now adjoin v_{j+1} to obtain the following family:

$$\{v_1, \dots, v_j, v_{j+1}, w_{j+1}, \dots, w_j\}$$

if this is linearly independent remove w_{j+1} and we are done. Else, some w_k is a linear combination of those preceding it by Lemma 1.5. Namely, there is $r > 0$ so that w_{j+r} is a linear combination of the preceding vectors. Remove the first such vector. Then, consider

$$\{v_1, \dots, v_j, v_{j+1}, w_{j+1}, \dots, \widehat{w_{j+r}}, \dots, w_m\} \quad (2)$$

We claim that this family is linearly independent. Else, it is linearly dependent and so we may write a non-trivial linear combination

$$\alpha_1 v_1 + \dots + \alpha_j v_j + \alpha_{j+1} v_{j+1} + \alpha_{j+1} w_{j+1} \dots + \alpha_{j+r} \widehat{w_{j+r}} + \dots + \alpha_m w_m = 0$$

observe that $\alpha_{j+1} \neq 0$, for this would contradict our induction hypothesis. Now, this implies that we may write v_{j+1} as a linear combination of the vectors above. Especially, we may substitute this into (2) to write w_{j+r} as a linear combination of the vectors in our original family: a contradiction. Hence, the lemma holds.

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It is useful to note that this algorithm above tells us *exactly* how to find the “culprit” vector.

Equipped with these notions, we may give the following theorem:

Theorem 1.7. *Every basis of a vector space V has the same cardinality. In particular, $\dim(V)$ is well defined.*

Proof. Let $\mathcal{B} := \{b_j\}_{j=1}^n$ be a basis for V and let $\mathcal{C} := \{c_j\}_{j=1}^m$ be another basis. We show $n = m$. First assume $m < n$. By Steinitz' substitution lemma above, we may “extend” this basis by adding elements of $\mathcal{B}$ to $\mathcal{C}$, but this is absurd since by Corollary 1.4 the family $\mathcal{C}$ is maximally independent. This proves that $m \not\leq n$ and a symmetric argument gives $m \not\geq n$. Hence, $m = n$ and $\dim(V) = n$.

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Especially, this argument tells us that if $\mathcal{S}$ is linearly independent then $\#\mathcal{S} \leq \dim(V)$. Moreover, if $\#\mathcal{S} = \dim(V)$ then $\mathcal{S}$ is a basis for V . Indeed, otherwise $\mathcal{S}$ is not maximally independent and one could adjoin an element to $\mathcal{S}$ whilst retaining independence, which we now know to be impossible. Finally, this process tells us that any independent set in V can be completed to a basis by successively applying Steinitz' lemma.

Proposition 1.8. *Let V be a vector space over $\mathbb{K}$ and W a subspace of V . Then, $W = V$ if and only if $\dim(W) = \dim(V)$.*

Proof. Assume that $W = V$, then clearly a basis for W is a basis for V and vice-versa. Therefore, $\dim(V) = \dim(W)$. Conversely suppose that $\dim(W) = \dim(V)$. Then, a basis $\mathcal{B}$ for W must be a basis for V since $\#\mathcal{B} = \dim(V)$ and this concludes the proof.

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1.3 LINEAR TRANSFORMATIONS

Let V and U be vector spaces over a field $\mathbb{K}$, a mapping $\mathbb{L} : U \rightarrow V$ is said to be a linear transformation of vector spaces (or homomorphism of vector spaces) if and only if each of the following hold true:

1. $\mathbb{L}(u + v) = \mathbb{L}(u) + \mathbb{L}(v)$ for all $u, v \in U$.
2. $\mathbb{L}(\alpha u) = \alpha \mathbb{L}(u)$ for all $u \in U$ and $\alpha \in \mathbb{K}$.

The following results are reminiscent of group theory. Given a homomorphism of vector spaces $\mathbb{L} : U \rightarrow V$ we define the **kernel** of $\mathbb{L}$, denoted $\ker \mathbb{L}$, to be the set consisting of all $u \in U$ mapping to 0 under $\mathbb{L}$. Namely, $\ker \mathbb{L} = \{u \in U : \mathbb{L}(u) = 0\}$. Similarly, $\Im(\mathbb{L}) = \{\mathbb{L}(u) : u \in U\}$ is a subset of V called the **image** of $\mathbb{L}$.

Lemma 1.9. *$\ker \mathbb{L}$ is a subspace of U .*

Proof. Obviously, if $u, v \in \ker \mathbb{L}$ then $\mathbb{L}(u + v) = \mathbb{L}(u) + \mathbb{L}(v) = 0 + 0 = 0$ and so $(u + v) \in \ker \mathbb{L}$. In a similar vein, if $u \in \ker \mathbb{L}$ then $\alpha u \mapsto 0$ and so $\alpha u \in \ker \mathbb{L}$.

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Lemma 1.10. *$\Im(\mathbb{L})$ is a subspace of V .*

Proof. If $u, v \in \Im(\mathbb{L})$ then there are $u', v' \in U$ so that $\mathbb{L}(u') = u$ and $\mathbb{L}(v') = v$. Hence, $(u' + v') \mapsto u + v$ proving that $u + v \in \Im(\mathbb{L})$ and a similar argument shows closure under scalar multiplication.

○

1.3.1 Matrix Representation

Here we give an alternative description of linear transformations $\mathbb{L}$. We claim that any $\mathbb{L}$ is given by a matrix and that any matrix represents some homomorphism of vector spaces. Let now V be a vector space of finite dimension n over a field $\mathbb{K}$ (sometimes it is helpful to think of $\mathbb{K}$ as $\mathbb{R}$ or $\mathbb{C}$) and fix a basis $\mathcal{B} = \{b_j\}_{j=1}^n$ for V . Then any vector $u \in V$ may be written (uniquely) as $\sum_{j=1}^n \alpha_j b_j$. It then makes sense to define the **coordinates** of u relative to $\mathcal{B}$ by

$$[u]_{\mathcal{B}} := \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}$$

Having equipped ourselves with this notion, it is now possible for us to describe linear transformations $\mathbb{L}$ by matrices.

Definition 5. Let V be a vector space over a field $\mathbb{K}$. An **endomorphism** of V is a linear map $\mathbb{L} : V \rightarrow V$.

We now turn our attention towards describing endomorphisms with matrices. Assume that $\mathcal{B} = \{b_j\}_{j=1}^n$ and $\mathcal{C} = \{c_j\}_{j=1}^n$ are two bases for a vector space V of finite dimension. Then, if we write the system:

$$\begin{cases} b_1 = m_{1,1}c_1 + m_{2,1}c_2 + \cdots + m_{n,1}c_n \\ b_2 = m_{1,2}c_1 + m_{2,2}c_2 + \cdots + m_{n,2}c_n \\ \vdots \\ b_n = m_{1,n}c_1 + m_{2,n}c_2 + \cdots + m_{n,n}c_n \end{cases} \quad (3)$$

then it is clear that the matrix

$${}_C M_{\mathcal{B}} = \begin{pmatrix} m_{1,1} & \cdots & m_{1,n} \\ \vdots & \ddots & \vdots \\ m_{n,1} & \cdots & m_{n,n} \end{pmatrix} \quad (4)$$

allows one to express a vector with respect to a different basis. Namely, $[u]_{\mathcal{C}} = {}_C M_{\mathcal{B}} [u]_{\mathcal{B}}$ for all $u \in V$. We have the following theorem about matrix endomorphisms:

Lemma 1.11. For all $u, v \in V$ and $\alpha \in \mathbb{K}$ one has $[u + v]_{\mathcal{B}} = [u]_{\mathcal{B}} + [v]_{\mathcal{B}}$ and $[\alpha u]_{\mathcal{B}} = \alpha [u]_{\mathcal{B}}$.

The lemma above is obvious and requires no proof. We are ready to prove the claim above

Proposition 1.12. *We have $[u]_C = {}_C M_B [u]_B$ for all $u \in V$.*

Proof. By linearity and scalar multiplication, it is sufficient to show that this holds for a basis element only. Namely, we need to show that:

$${}_C M_B [b_j]_B = {}_C M_B e_j = \text{j-th column of } {}_C M_B = [b_j]_C$$

○

Lemma 1.13. *The matrix ${}_C M_B$ above is uniquely determined for fixed bases.*

Proof. Assume M is a matrix so that $[u]_C = {}_C M_B [u]_B$ for all $u \in V$. Note that the j-th column of ${}_C M_B$ corresponds to $[b_j]_C = M[b_j]_B = M e_j$ which is the j-th column of M . Hence, $M = {}_C M_B$.

○

We now have the following results:

Corollary 1.14.

1. ${}_B M_B = I_n$, the $n \times n$ identity matrix.
2. ${}_{\nabla} M_B = {}_{\nabla} M_C {}_C M_B$.
3. The matrix ${}_C M_B$ has inverse ${}_B M_C$.

Proof.

1. This follows from the previous lemma. Certainly, it follows that $[u]_B = I_n [u]_B$ for all $u \in V$ and therefore by uniqueness we have (1).
2. Again follows immediately from the previous lemma.
3. We prove this by using (1)-(2). Certainly, note that from (2) we know that ${}_B M_C \cdot {}_C M_B = I_n$ for $\dim(V) = n$. Thence, we deduce that each of these matrices are invertible and are their own respective inverses.

○

1.3.2 Isomorphisms of Vector Spaces

This is the more interesting side of linear mappings. A linear transformation (or homomorphism of vector spaces) $L : U \rightarrow V$ is said to be an isomorphism if and only if L is *bijective*. Recall that a bijective mapping is a function that is one-one: every element in V has exactly one point in U mapping to it. We say two vector spaces U, V over a field $\mathbb{K}$ are *isomorphic* provided there exists an isomorphism $L : U \rightarrow V$ between them. In which case, we would write $U \cong V$.

This is an equivalence relation on vector spaces. Indeed, the identity mapping is an isomorphism from a vector space to itself and so $U \cong U$ for all vector spaces U . Since a bijection is invertible, the inverse of an isomorphism is again an isomorphism, thence $U \cong V$ if and only if $V \cong U$. Finally, since the composition of isomorphisms is again an isomorphism we have that if $U \cong V$ and $V \cong W$ that $U \cong W$.

Theorem 1.15. *Let V be a vector space of dimension $n < \infty$ over a field $\mathbb{K}$. Then, $V \cong \mathbb{K}^n$.*

Proof. Consider now the mapping $\mathbb{L} : V \rightarrow \mathbb{K}^n$ given by $u \mapsto [u]_{\mathcal{B}}$ for a basis $\mathcal{B} = \{b_j\}_{j=1}^n$ for V . This is clearly a linear mapping: i.e. a homomorphism of vector spaces. Now, we need only show that this is a bijection. Surjectivity is clear from the fact that the vector $v := \sum_{j=1}^n \alpha_j b_j \mapsto (\alpha_j)_{j=1}^n \in \mathbb{K}^n$. As for injectivity, we need only show that $\ker \mathbb{L} = \{0\}$. This is obvious since having all coordinates 0 implies that $u \equiv 0$. Hence, $V \cong \mathbb{K}^n$. ○

Theorem 1.16. *Let $\mathbb{L} : U \rightarrow V$ be an isomorphism of vector spaces. If $\mathcal{B}$ is a basis for U , then $\mathbb{L}(\mathcal{B})$ is a basis for V .*

Proof. Let $\mathcal{B} = \{b_j\}_{j=1}^n$ be a basis for U , we must show that $\{\mathbb{L}(b_j)\}_{j=1}^n$ is a basis for V . For this, it is sufficient to show that it is maximally independent. To see that it is independent, observe that if $\sum_{j=1}^n \alpha_j \mathbb{L}(b_j) = 0$ then $\mathbb{L}(\sum_{j=1}^n \alpha_j b_j) = 0$ implying that $\sum_{j=1}^n \alpha_j b_j = 0$ since $\ker \mathbb{L} = \{0\}$. But $\mathcal{B}$ is a basis and is therefore independent whence we find that $\alpha_j = 0$ for all indices j .

It remains to show that $\{\mathbb{L}(b_j)\}_{j=1}^n$ is maximally independent. Let $v \in V$ be given. Then, since $\mathbb{L}$ is a surjection there exists a combination $\sum_{j=1}^n \beta_j b_j$ mapping to v under $\mathbb{L}$. It follows that $\{\mathbb{L}(b_1), \mathbb{L}(b_2), \dots, \mathbb{L}(b_n), v\}$ is linearly dependent because v is a linear combination of the vectors preceding it. The theorem is now proven. ○

Theorem 1.17. *Let U, V be vector spaces over $\mathbb{K}$ and $\mathbb{L} : U \rightarrow V$ a linear transformation. Then,*

$$\dim(U) = \dim(\ker \mathbb{L}) + \dim(\Im(\mathbb{L})) \quad (5)$$

Proof. Let $\{b_j\}_{j=1}^n$ be a basis for $\ker \mathbb{L}$, which we know to be a subspace of U . An application of Lemma 1.6 tells us that we may adjoin elements $\{c_1, \dots, c_m\}$ to turn this collection into a basis for the “whole” space U . We are done if we can show that $\{\mathbb{L}(c_1), \dots, \mathbb{L}(c_m)\}$ is a basis for $\Im(\mathbb{L})$. We shall show that this collection is minimally spanning, whence the result follows from Corollary 1.4.

We first show that this is spanning. Indeed, let $v \in \Im(U) \subseteq V$. Then there is a point $u \in U$ so that $u \xrightarrow{\mathbb{L}} v$. Write now $u = \sum_{j=1}^n \alpha_j b_j + \sum_{j=1}^m \beta_j c_j$. Now, by linearity we have that $\mathbb{L}(u) = \sum_{j=1}^m \beta_j \mathbb{L}(c_j) = v$ implying that this is a spanning set.

We now show that it is minimal. Certainly, let us remove some element of this collection, after a reordering we may suppose that this element was $\mathbb{L}(c_m)$. If $\{\mathbb{L}(c_j)\}_{j=1}^{m-1}$ is still a spanning set then we may write $\mathbb{L}(c_m) = \sum_{j=1}^{m-1} \gamma_j \mathbb{L}(c_j)$. Now, by linearity we have

$$\mathbb{L} \left(\sum_{j=1}^{m-1} \gamma_j c_j - c_m \right) = 0$$

Therefore this combination lives in $\ker \mathbb{L}$ and therefore we may write $\sum_{j=1}^{m-1} \gamma_j c_j - c_m = \sum_{i=1}^k \delta_i b_i$. But this is a non-trivial combination that contradicts the linear independence of the constructed basis for U .

○

This theorem is oft called the *theorem about the kernel and the image*; it is a powerful tool and we shall now use it to deduce many results about vector spaces.

Corollary 1.18. *Let $\mathbb{L} : U \rightarrow V$ be a linear transformation and assume that $\dim(U) = \dim(V)$. Then*

1. *If $\mathbb{L}$ is injective it is an isomorphism of vector spaces.*
2. *If $\mathbb{L}$ is surjective it is an isomorphism of vector spaces.*

Proof.

1. Assume that $\ker \mathbb{L} = \{0\}$, then $\dim(\ker \mathbb{L}) = 0$ and $\dim(U) = \dim(\Im(\mathbb{L})) = \dim(V)$. Now, this implies that $\Im(\mathbb{L}) = V$ and hence $\mathbb{L}$ is an isomorphism (since it is surjective).
2. This is a similar argument. We have that $\dim(U) = \dim(V) = \dim(\ker \mathbb{L}) + \dim(\Im(\mathbb{L}))$ which implies that $\dim(\ker \mathbb{L}) = 0$ and so $\ker \mathbb{L} = \{0\}$.

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1.3.3 Operators and Projections

1.3.3.1 Inner Direct Summation

We now consider a family of vector subspaces: let V be a vector space over a field $\mathbb{K}$ and consider a family of subspaces $\{U_j\}_{j=1}^n$ of V . Assume further that

1. $U_1 + U_2 + \cdots + U_n = V$.
2. For all j one has $U_j \cap (U_1 + \cdots + \widehat{U_j} + \cdots + U_n) = \emptyset$.

We now call V the *inner direct product* of the subspaces U_j and may write $V = \bigoplus_{j=1}^n U_j$.

Lemma 1.19. $V = \bigoplus_{j=1}^n U_j$ if and only if the mapping

$$\mathbb{L} : \bigoplus_{j=1}^n U_j \rightarrow V, \quad (u_1, u_2, \dots, u_n) \mapsto \sum_{j=1}^n u_j$$

is an isomorphism of vector spaces.

1.3.3.2 Nilpotent Operators

We remark that an endomorphism $\mathbb{L} : V \rightarrow V$ is often called an operator. We shall say an endomorphism $\mathbb{L} : V \rightarrow V$ is *nilpotent* if there exists $N \in \mathbb{N}$ so that $\mathbb{L}^N \equiv 0$. A simple result is the following

Lemma 1.20. *Let V be an n -dimensional vector space and $\mathbb{L}$ a nilpotent operator on a vector space V . Then $\mathbb{L}^n \equiv 0$.*

Proof. Set $V_0 := V$ and $V_1 := \mathbb{L}(V_0)$. For all $i \in \mathbb{N}$ we let $V_i := \mathbb{L}(V_i)$. We claim for all $i \in \mathbb{N}$ one has $\dim(V_i) > \dim(V_{i+1})$. Certainly, by a previous lemma we know that V_{i+1} is a subspace of V_i and therefore we only have strict equality in the dimensions if $V_i = V_{i+1}$. But then for all $j \geq i$ one would have $V_j = V_{j+1}$ which is a contradiction.

○

We shall return to the study of linear transformations later on, and describe these as matrices just as before.

MATRICES

In this section we discuss the topic of matrices. We have already seen in the previous chapter that we may express endomorphisms of finite-dimensional vector spaces via matrices, we shall see later on that we may express most homomorphisms with matrices. Naturally, it makes sense to define the ring of all matrices over a field $\mathbb{K}$. Certainly, we shall denote by $M_n(\mathbb{K})$ the *ring* of all $n \times n$ matrices over a field $\mathbb{K}$. Observe with care that this is not a commutative ring, and certainly not a division ring.

2.1 CONSTRUCTION OF THE DETERMINANT TENSOR

We recall the existence of a surjective homomorphism of groups $\text{sgn} : S_n \rightarrow \{\pm 1\}$ where S_n is the group of permutations on n -letters. Recall that a permutation is a function $\sigma : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ that is bijective.

Now we may construct this function $\det(\cdot) : M_n(\mathbb{K}) \rightarrow \mathbb{K}$, a tensor, which we all know and love. In our construction we will use $v_1, v_2, \dots, v_n$ to denote column vectors in $\mathbb{K}^n$ and will write $(v_1 \mid v_2 \mid \dots \mid v_n)$ to describe a matrix in $M_n(\mathbb{K})$ with columns consisting of these vectors. Let us begin with a definition:

Definition 6. Let U, V be vector spaces over $\mathbb{K}$ and assume $T : U^k \rightarrow V^k$ is a mapping. We shall call T *multi-linear* provided that

$$T(u_1, u_2, \dots, \alpha u_j, \dots, u_n) = \alpha T(u_1, u_2, \dots, u_n) \quad (6)$$

$$T(u_1, \dots, u_j + u'_j, \dots, u_n) = T(u_1, \dots, u_j, \dots, u_n) + T(u_1, \dots, u'_j, \dots, u_n) \quad (7)$$

for all indices $1 \leq j \leq n$.

Theorem 2.1. Fix a field $\mathbb{K}$ and let $n \in \mathbb{N}$ be given. There exists a multi-linear function

$$\det(\cdot) : M_n(\mathbb{K}) \rightarrow \mathbb{K} \quad (8)$$

Such that $\det(I_n) = 1$ and $\det(A) = 0$ if A has two duplicate columns.

Proof. We construct the function as follows. Define a mapping $\det(\cdot) : M_n(\mathbb{K}) \rightarrow \mathbb{K}$ by letting:

$$\boxed{\det(A) := \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}, \quad A = (a_{i,j})_{i,j} \in M_n(\mathbb{K})} \quad (\dagger)$$

In the above i describes a row and j defines a row. It suffices to show that this is a multilinear mapping taking I_n to 1 and vanishes whenever columns are duplicated.

We begin by showing linearity. Assume we have a matrix $(u_1 \mid u_2 \mid \dots \mid u_n) = (a_{i,j})_{i,j}$ and we consider the matrix $(b_{i,j})$ corresponding to $(u_1 \mid \dots \mid \alpha u_j \mid \dots \mid u_n)$. We claim now that this simply “multiplies” the determinant of $(a_{i,j})_{i,j}$. Certainly, for $\alpha \in \mathbb{K}$:

$$\det(u_1 \mid \dots \mid \alpha u_j \mid \dots \mid u_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) b_{\sigma(1),1} \cdots b_{\sigma(n),n}$$

Now, σ is a permutation on n -letters and is therefore a bijection; implying that in any summand there is exactly one factor of α , whence we see that the above essentially reduces to

$$\begin{aligned} \det(u_1 \mid \dots \mid \alpha u_j \mid \dots \mid u_n) &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) b_{\sigma(1),1} \cdots b_{\sigma(n),n} = \alpha \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n} \\ &= \alpha \det(u_1 \mid \dots \mid u_n) \end{aligned}$$

Now, we shall finish the proof of multilinearity. Let $u_j, u'_j \in \mathbb{K}^n$ be column vectors, and consider now the matrix $(b_{i,j})_{i,j}$ given by $(u_1 \mid \dots \mid u_j + u'_j \mid \dots \mid u_n)$. Then this matrix has the following determinant:

$$\det((b_{i,j})_{i,j}) = \det(u_1 \mid \dots \mid u_j + u'_j \mid \dots \mid u_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) b_{\sigma(1),1} \cdots b_{\sigma(n),n}$$

But this last line corresponds to $\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots (a_{\sigma(j),j} + a'_{\sigma(j),j}) \cdots a_{\sigma(n),n}$ which as we can tell is simply the sum

$$\begin{aligned} \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(j),j} \cdots a_{\sigma(n),n} + \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a'_{\sigma(j),j} \cdots a_{\sigma(n),n} \\ = \det(u_1 \mid \dots \mid u_j \mid \dots \mid u_n) + \det(u_1 \mid \dots \mid u'_j \mid \dots \mid u_n) \end{aligned}$$

Which proves that $\det(\cdot)$ is a multi-linear mapping. It is useful to observe that in particular we have that if a column consists entirely of zeros, then the determinant vanishes. Indeed, let $A := (u_1 \mid \dots \mid u_\ell \mid \dots \mid u_n)$ where u_ℓ is a column vector consisting entirely of zeros. Then from the linearity in the ℓ -th column it is easy to deduce that

$$\det(A) = \det(u_1 \mid \dots \mid u_\ell \mid \dots \mid u_n) = \det(u_1 \mid \dots \mid u_\ell + u_\ell \mid \dots \mid u_n)$$

and therefore that

$$\det(u_1 \mid \dots \mid u_\ell \mid \dots \mid u_n) = \det(u_1 \mid \dots \mid u_\ell \mid \dots \mid u_n) + \det(u_1 \mid \dots \mid u_\ell \mid \dots \mid u_n)$$

whence $\det(A) = \det(A) + \det(A)$ implying that $\det(A) = 0$.

Now we return to the construction. We shall denote by $I_n \in M_n(\mathbb{K})$ the *identity matrix* of $n \times n$ dimensions. Namely, we shall write

$$I_n := (e_1 \mid e_2 \mid \cdots \mid e_n) = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \quad (9)$$

We shall establish that $\det(I_n) = 1$, for all $n \in \mathbb{N}$. Consider now the Kronecker delta:

$$\delta_{i,j} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

It is clear that we may then express $I_n = (\delta_{i,j})_{i,j}$ for $1 \leq i, j \leq n$. Now, we employ the definition of $\det(\cdot)$ as follows:

$$\det(I_n) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \delta_{\sigma(1),1} \delta_{\sigma(2),2} \cdots \delta_{\sigma(n),n} \quad (10)$$

The important observation from equation (10) is that a summand vanishes if and only if σ is not the trivial permutation: $\sigma(j) = j$ for all $j \in \{1, \dots, n\}$. Indeed, this is clear from the very definition of the Kronecker delta. By this we simply mean that if $\sigma(j) \neq j$ for some j then the summand for this σ is 0. Therefore, if we let τ describe the trivial permutation on n -letters the above summation reduces to

$$\det(I_n) = \text{sgn}(\tau) \delta_{\tau(1),1} \cdots \delta_{\tau(n),n} = \text{sgn}(\tau) \prod_{j=1}^n \delta_{j,j} = \text{sgn}(\tau)$$

but no “switches” are made, and so $\text{sgn}(\tau) = 1$ (the trivial permutation has positive sign). Hence we have found that $\det(I_n) = 1$.

Now, to prove the following property assume that $u_j = u_i$ for $i < j$. Then, we need only calculate $\det(u_1 \mid \cdots \mid u_i \mid \cdots \mid u_j \mid \cdots \mid u_n)$. We may now fix the indices i, j and consider the subgroup of S_n defined by $\{1, (ij)\} < S_n$. The reader may verify that this is indeed a subgroup, and this is clear from the fact that $(ij)(ij) = 1$. We may fix a set of representatives $S \subseteq S_n$ for the aforementioned subgroup; then $S_n = S \sqcup S(ij)$ where $S(ij)$ are right cosets. Thence, since this union is *disjoint* we find that

$$\begin{aligned} \det(u_1 \mid \cdots \mid u_i \mid \cdots \mid u_j \mid \cdots \mid u_n) &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n} \\ &= \sum_{\sigma \in S} \text{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n} + \sum_{\varsigma \in S(ij)} \text{sgn}(\varsigma) a_{\varsigma(1),1} \cdots a_{\varsigma(n),n} \end{aligned}$$

where $\varsigma = \sigma(ij)$ for some pair (ij) . However, then the above simply reduces to

$$\begin{aligned} \det(u_1 \mid \cdots \mid u_i \mid \cdots \mid u_j \mid \cdots \mid u_n) &= \sum_{\sigma \in S} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n} \\ &\quad + \sum_{\varsigma \in S(ij)} \operatorname{sgn}(\varsigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n} \end{aligned}$$

and since $\operatorname{sgn}(\varsigma) = -\operatorname{sgn}(\sigma)$ this determinant vanishes. The proof of this theorem is now complete. ○

2.2 GEOMETRIC INTERPRETATION OF THE DETERMINANT MAP

This last section was particularly abstract, and I'll admit that the definition of the determinant given is far from intuitive. Therefore, it makes sense for us to present a section in which we attempt to give the reader a more intuitive and familiar understanding of this multi-linear mapping from $M_n(\mathbb{K}) \rightarrow \mathbb{K}$.

Many of the readers may be familiar with the determinant as a *signed volume function*. Namely, it returns the volume (up to a sign) of a parallelepiped spanned by the vectors $(u_j)_{j=1}^n$, i.e. the edges of the figure are given by these vectors.

Now, this does not exactly tell us how to compute the determinant and this has yet to be justified. By way of justification, consider a one dimensional matrix: i.e. a scalar. Then $\det(a)$ is simply the length of the segment from 0- a upto a sign; $\det(a) = a \in \mathbb{K}$. In the case of a 2×2 matrix, we have a somewhat more difficult calculation to make.

Consider the matrix M given by $\begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix}$. We wish to compute the determinant of this matrix, in the hopes of having an easy access formula for "smaller matrices". The key step is to observe that the set of permutations on 2-letters is rather small, and is given by $S_2 := \{1, (12)\}$. It follows from this that

$$\begin{aligned} \det \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} &= \sum_{\sigma \in S_2} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} = a_{1,1} a_{2,2} + \operatorname{sgn}((12)) a_{\sigma(1),1} a_{\sigma(2),2} \\ &= a_{1,1} a_{2,2} - a_{2,1} a_{1,2} \end{aligned} \tag{*1}$$

The equation in $(\star_1)$ is particularly useful in practice. For instance, when using a change of variables it is very useful in the computation of the jacobian matrix. In practice, one does not compute the determinant via the definition, but rather from consequential formulae that follows. We shall see many ways of computing these later on.

2.3 MULTIPLICATIVITY OF THE DETERMINANT

In this section we explore one of the crucial computational tools with the determinant: *multiplicativity*. Namely, we seek to prove here that $\det(AB) = \det(A) \det(B)$ for all elements $A, B \in M_n(\mathbb{K})$. In practice this will allow us to easily compute complex determinants with ease. It will allow us to develop more complex tools such as *Laplace's theorem*.

Theorem 2.2. *For all $A, B \in M_n(\mathbb{K})$ we have $\det(AB) = \det(A) \det(B)$.*

We do not yet have all the tools to prove this theorem. In fact, deducing this from the definition of the determinant is *quite tricky*, and we shall take an alternative approach to establishing this fact. First, we have:

Lemma 2.3. *Let $\mathcal{D} : M_n(\mathbb{K}) \rightarrow \mathbb{K}$ be a multilinear map (with respect to column vectors) such that $\mathcal{D}(A) = 0$ if any two columns of A are duplicates. Then, $\mathcal{D}(A) = \xi \det(A)$ where $\xi = \mathcal{D}(I_n)$ for all $A \in M_n(\mathbb{K})$.*

Proof. Consider now a collection of column vectors $\{u_j\}_{j=1}^n$, and write $u_j = \sum_{i=1}^n a_{ij}e_i$ where e_i 's form the standard basis in $\mathbb{K}^n$. In this case, we have expressed the matrix

$$(u_1 \mid \cdots \mid u_n) = (a_{ij})_{i,j}$$

Now we simply use the (multi)-linearity of this function ∇ :

$$\begin{aligned} \nabla(A) &= \nabla(u_1 \mid \cdots \mid u_n) = \nabla\left(\sum_{i=1}^n a_{i1}e_i \mid \cdots \mid \sum_{i=1}^n a_{in}e_i\right) \\ &= \sum_{(i_1, \dots, i_n)} a_{i_1} \cdots a_{i_n} \mathcal{D}(e_{i_1} \mid \cdots \mid e_{i_n}) \end{aligned}$$

By assumption on $\mathcal{D}$ we find that the only summands that are non-zero are those where all the i_k are distinct. But this implies that the collection $(i_k)_{k=1}^n$ is in fact a *permutation* on n -letters, i.e. corresponds to some $\sigma \in S_n$. Therefore,

$$\begin{aligned} \mathcal{D}(A) &= \sum_{\sigma \in S_n} a_{\sigma(1),1} \cdots a_{\sigma(n),n} \mathcal{D}(e_{\sigma(1)} \mid \cdots \mid e_{\sigma(n)}) \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n} \mathcal{D}(e_1 \mid \cdots \mid e_n) \\ &= \xi \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n} \\ &= \xi \det(A) \end{aligned}$$

which concludes the proof. ○

This lemma lends itself to a powerful corollary

Corollary 2.4. *The function $\det(\cdot) : M_n(\mathbb{K}) \rightarrow \mathbb{K}$ is the unique multilinear mapping such that $\det(I_n) = 1$ and $\det(A) = 0$ if columns are duplicated.*

Proof. By the above if $\mathcal{D}(\cdot) : M_n(\mathbb{K}) \rightarrow \mathbb{K}$ is another candidate function we have $\mathcal{D}(A) = \det(A) \cdot \mathcal{D}(I_n)$ for all $A \in M_n(\mathbb{K})$. But since $\mathcal{D}(I_n) = 1$ we find that $\det(\cdot) = \mathcal{D}(\cdot)$ on $M_n(\mathbb{K})$. ○

Proof of Theorem 2.2. By corollary 2.4 it suffices to show that the mapping $\mathcal{D}(\cdot) : M_n(\mathbb{K}) \rightarrow \mathbb{K}$ given by $\mathcal{D}(\cdot) := \det(A \cdot)$ satisfies the properties of the determinant. We begin by defining some notation. Suppose we are given two vectors, in column and row form:

$$\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, \quad (v_1, v_2, \dots, v_n) \quad (11)$$

Then we may define a mapping $\langle \cdot, \cdot \rangle : \mathbb{K}^n \times \mathbb{K}^n \rightarrow \mathbb{K}$ by $\langle u, v \rangle = \sum_{j=1}^n u_j v_j$. As we shall see, this is the *dot-product* and is an inner-product, and turns $\mathbb{K}^n$ into an *inner-product space*. For now, this just gives us a convenient way to describe matrix multiplication. Indeed, if we write

$$A = \begin{pmatrix} u_1 \\ \text{---} \\ \vdots \\ \text{---} \\ u_n \end{pmatrix}, \quad B = (v_1 \mid \dots \mid v_n)$$

then we may express the product¹ $AB \in M_n(\mathbb{K})$ as:

$$AB = \begin{pmatrix} \langle u_1, v_1 \rangle & \cdots & \langle u_1, v_n \rangle \\ \vdots & \ddots & \vdots \\ \langle u_n, v_1 \rangle & \cdots & \langle u_n, v_n \rangle \end{pmatrix} \quad (\star_2)$$

Using the expression in $(\star_2)$ we shall show that it satisfies the desired determinant properties, and conclude from Lemma 2.3 that $\det(AB) = \det(A) \det(B)$. Define $\mathcal{D} : M_n(\mathbb{K}) \rightarrow \mathbb{K}$ by taking $M_n(\mathbb{K}) \ni B \mapsto \det(AB)$ where $A \in M_n(\mathbb{K})$ is fixed. The linearity of $\mathcal{D}$ follows immediately from the vector space structure of $M_n(\mathbb{K})$ over $\mathbb{K}$.

¹ Equation $(\star_2)$ gives us a convenient and easy to remember closed form for the matrix product.

Now, it is clear from ($\star_2$) that if $v_j = v_\ell$ for $\ell \neq j$ then two columns of AB are equal, and hence $\mathcal{D}$ vanishes by our construction.

Now, since $\mathcal{D}(I_n) = \det(A)$ we find from Lemma 2.3 that $\det(AB) = \det(A) \det(B)$ as was asserted. ○

2.4 THE ADJOINT MATRIX

Again, this section gives us insight on the determinant and provides “tricks” to compute determinants and inverses² of matrices, when the latter exists. Recall from the very definition in ($\dagger$) that given a matrix $A \in M_n(\mathbb{K})$ the determinant of A is given by

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} \cdots a_{\sigma(n),n} \quad (12)$$

From this we may find that for a fixed index j we have the alternative formula

$$\det(A) = \sum_{i=1}^n a_{i,j} \sum_{\{\sigma: \sigma(j)=i\}} \operatorname{sgn}(\sigma) \prod_{\ell \neq j} a_{\sigma(\ell),\ell} \quad (13)$$

Definition 7 (Minor and Cofactors). Let $A = (a_{i,j})$ be a matrix in a field $\mathbb{K}$. For a pair $(i, j) \in \{1, \dots, n\}^2$ we define the i, j -minor of A , denoted $A_{i,j}$ to be the matrix obtained from A by deleting the i -th row and j -th column and the i, j -th cofactor to be $(-1)^{i+j} \det(A_{i,j})$ which we oft denote $A^{i,j}$.³

We may the observation that from our definition in ($\dagger$)

$$\det(A) = \sum_{i=1}^n a_{i,j} \sum_{\{\sigma: \sigma(j)=i\}} \operatorname{sgn}(\sigma) \prod_{\ell \neq j} a_{\sigma(\ell),\ell}, \quad A = (a_{i,j}) \in M_n(\mathbb{K}) \quad (\star_3)$$

The motivating result is the following:

Proposition 2.5. $\sum_{\zeta: \zeta(j)=i} [\operatorname{sgn}(\zeta) \prod_{\ell \neq j} a_{\zeta(\ell),\ell}] = A^{i,j}$.

This is a result from the theory of permutations of groups and we shall omit the proof. This proposition does give us a powerful tool due to Laplace:

Theorem 2.6 (Laplace). Let $A \in M_n(\mathbb{K})$ for $n \in \mathbb{N}$ and write $A = (a_{i,j})_{i,j}$. For any $1 \leq i, j \leq n$ one has the following expansions:

$$\det(A) = a_{i,1} A^{i,1} + a_{i,2} A^{i,2} + \cdots + a_{i,n} A^{i,n} = \sum_{\ell=1}^n a_{i,\ell} A^{i,\ell} \quad (14)$$

² This also gives us a sufficient and necessary condition to ensure matrix invertability.

³ Note that this is well defined because $A_{i,j} \in M_{n-1}(\mathbb{K})$

where we “develop by rows”. Developing by columns we find the analagous formula:

$$\det(A) = a_{1,j}A^{1,j} + a_{2,j}A^{2,j} + \cdots + a_{n,j}A^{n,j} = \sum_{\ell=1}^n a_{\ell,j}A^{\ell,j} \quad (15)$$

Moreover, if $\ell \neq j$ then $\sum_{i=1}^n a_{i,j}A^{i,\ell} = 0$ and similarly if $\ell \neq i$ then $\sum_{j=1}^n a_{i,j}A^{\ell,j} = 0$.

Before we prove this theorem let us illustrate it with a practical example. Consider the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \in M_3(\mathbb{R})$. Directly from the definition of the determinant in (†) it would be quite difficult to compute $\det(A)$, but Theorem 2.6 gives us an efficient way of doing so. Certainly, we expand along the first row:

$$\det(A) = \det \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = 1 \cdot \det \begin{pmatrix} 5 & 6 \\ 8 & 9 \end{pmatrix} + (-1)2 \cdot \det \begin{pmatrix} 4 & 6 \\ 7 & 9 \end{pmatrix} + 3 \det \begin{pmatrix} 4 & 5 \\ 7 & 8 \end{pmatrix}$$

which we are already familiar with. We now give the following lemma

Lemma 2.7. *Let $A \in M_n(\mathbb{K})$. Then, $\det(A^T) = \det(A)$.*

Proof. We simply employ the definition of the determinant in the case of $B := A^T$ which we denote by $(b_{i,j})$. The key observation to make here is that for any two permutations $\sigma, \varsigma \in S_n$ one has

$$\prod_{j=1}^n a_{\sigma(j),j} = \prod_{j=1}^n a_{\sigma(\varsigma(j)),\varsigma(j)}$$

Indeed, this follows from the fact that ς is a bijection on n -letters. In particular, the above holds when we take $\varsigma = \sigma^{-1}$; again a permutation. Therefore applying (†) to A we find the following:

$$\begin{aligned} \det(A) &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j} = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(\sigma^{-1}(j)),\sigma^{-1}(j)} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{j,\sigma^{-1}(j)} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma^{-1}) \prod_{j=1}^n a_{j,\sigma^{-1}(j)} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\varsigma) \prod_{j=1}^n a_{j,\varsigma(j)} \\ &= \sum_{\sigma \in S_n} \operatorname{sgn}(\varsigma) \prod_{j=1}^n b_{\varsigma(j),j} \end{aligned}$$

which is precisely the definition of $\det(B) = \det(A^T)$ which completes the proof.

○

Proof of Theorem 2.6. Observe that (15) is immediate from Proposition 2.5 together with (★₃). We have from Lemma 2.7 that $\det(A) = \det(A^T)$ whence one sees (14). Now suppose that $\ell \neq j$. We shall show that $\sum_{i=1}^n a_{i,j} A^{i,\ell} = 0$. This is easily seen by replacing the ℓ -th column of the matrix by the j -th column. This does not affect the co-factors $A^{i,\ell}$ but changes the $a_{i,\ell}$ to $a_{i,j}$. Therefore the expression above is precisely the determinant of this new matrix, which is 0 because two columns are identical. This completes the proof.

○

Definition 8 (Adjoint Matrix). Let $A = (a_{ij}) \in M_n(\mathbb{K})$. We define a new matrix denoted $\text{Adj}(A) := (c_{ij})$ where $c_{ij} = A^{j,i}$. In other words $\text{Adj}(A)$ is the transpose of the cofactor matrix, which has ij -th entry equal to $A^{i,j}$.

Theorem 2.8. Let $A \in M_n(\mathbb{K})$ be given. Then,

$$\text{Adj}(A) \cdot A = A \cdot \text{Adj}(A) = \det(A) \cdot I_n \quad (\star_4)$$

Proof. We shall only prove the first equality here, leaving the second as an exercise in which a near verbatim argument applies. Now, consider the ij -th entry of the product in the matrix product $\text{Adj}(A) \cdot A$. Recalling our closed form in (★₂) we may easily calculate this entry:

$$(\text{Adj}(A) \cdot A)_{ij} = \sum_{\ell=1}^n (\text{Adj}(A))_{i,\ell} \cdot a_{\ell,j} = \sum_{\ell=1}^n a_{\ell,j} A^{\ell,i} = \begin{cases} \det(A) & i = j \\ 0 & i \neq j \end{cases}$$

○

A powerful criterion for invertability follows from the above theorem:

Theorem 2.9. Let $A \in M_n(\mathbb{K})$. Then A is invertible if and only if $\det(A) \neq 0$. In this case, we have that $A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$.

Proof. Assume that A is invertible. There exists some matrix $B \in M_n(\mathbb{K})$ such that $AB = I_n$, whence by Theorem 2.2 we see $1 = \det(AB) = \det(A) \det(B)$. Since $\mathbb{K}$ is an integral domain, we have that $\det(B) = \frac{1}{\det(A)}$ and hence $\det(A)$ is non-zero. Conversely, suppose that $\det(A) \neq 0$: i.e $\frac{1}{\det(A)}$ exists in $\mathbb{K}$. Then, we employ (★₄) to find

$$A \cdot \text{Adj}(A) = \text{Adj}(A) \cdot A = \det(A) \cdot I_n$$

Since $\det(A)^{-1}$ exists we may multiply through to find that $A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$ as was asserted.

○

2.4.1 Matrix Representations of Homomorphisms

Here we generalize the results we obtained for endomorphisms. Recall from the first chapter that we developed a method for representing endomorphisms $\mathbb{L} : V \rightarrow V$ as a *matrix*. Here, we shall repeat the same ideas but instead for homomorphisms of vector spaces $\mathbb{L} : U \rightarrow V$.

We begin with a useful theorem:

Theorem 2.10.

Let U, V be vector spaces over a field $\mathbb{K}$ and let $\mathbb{L} : U \rightarrow V$ a homomorphism of vector spaces. Let $\mathcal{B}$ is a basis for U and $\mathcal{C}$ a basis for V . There exists a unique matrix ${}_C[\mathbb{L}]_{\mathcal{B}}$ such that

$$[Lu]_{\mathcal{C}} = {}_C[\mathbb{L}]_{\mathcal{B}}[u]_{\mathcal{B}}$$

The construction is straightforward. Let enumerate the basis $\mathcal{B} = \{b_j\}_{j=1}^n$ and let the matrix M consist of the column vectors $\mathbb{L}(b_j)$ for all j relative to the basis $\mathcal{C}$.

SYSTEMS OF LINEAR EQUATIONS

In this section we shall (finally) tackle the problem most introductory linear algebra courses handle: solving systems of equations. This has been postponed until now for good reason: we have already developed most of the tools that will be needed to systematically attack these problems.

3.1 THE DUAL VECTOR SPACE AND DIMENSION

We now return to the study of a vector space. Again, suppose that V is a vector space over a field $\mathbb{K}$. There is a naturally induced vector space *dual* to V that arises from V . We first recall some definitions.

Definition 9. Let V be a vector space over a field $\mathbb{K}$. A functional F on V is a map $F : V \rightarrow \mathbb{K}$.

Now we are ready to define the dual of a vector space V .

Definition 10. Let V be a vector space over $\mathbb{K}$, we define the dual space on V , denoted V^* , as follows:

$$V^* := \{ \mathbb{L} : V \rightarrow \mathbb{K} : \mathbb{L} \text{ is a linear functional} \} \quad (16)$$

It is left to the reader that V^* forms a vector space over $\mathbb{K}$ in its own right. We shall instead turn our attention towards the following proposition:

Proposition 3.1. Let U, V be vector spaces and $\mathcal{B} := \{b_j\}_{j=1}^n$ a basis for U . Given a list $\{v_j\}_{j=1}^n \subseteq V$ there is a unique linear mapping $\mathbb{L} : U \rightarrow V$ so that $\mathbb{L}(b_j) = v_j$ for all j .

Proof. The existence is easy, simply set $\mathbb{L}(b_j) = v_j$ and induce a linear mapping on the entire space. This can certainly be done because $\mathcal{B}$ is a basis for U and hence any vector $u \in U$ is a linear combination of basis elements.

We shall now show uniqueness of this mapping. Assume $T : U \rightarrow V$ is another such mapping. Then any $u \in U$ has a unique expression as $u = \sum_{j=1}^n \alpha_j b_j$. Using linearity of $\mathbb{L}$ and T together with their agreement on $\mathcal{B}$ one sees that

$$\mathbb{L} \left(\sum_{j=1}^n \alpha_j b_j \right) = \sum_{j=1}^n \alpha_j \mathbb{L}(b_j) = \sum_{j=1}^n \alpha_j T(b_j) = T \left(\sum_{j=1}^n \alpha_j b_j \right)$$

which proves uniqueness. ○

This may be applied to obtain the following theorem:

Theorem 3.2. *Let V be a finite-dimensional vector space over a field $\mathbb{K}$. Then $\dim(V) = \dim(V^*)$.*

Proof. Since $\dim(V) = n < \infty$ we may extract a basis $\mathcal{B} = \{b_j\}_{j=1}^n$ for V . Now we construct elements of V^* from these basis elements as follows: by the preceding proposition there exists a unique linear mapping $f_i : V \rightarrow \mathbb{K}$ so that $f_i(b_j) = \delta_{ij}$ for all basis elements $b_j \in \mathcal{B}$.

Following the proof in this proposition we observe that these induce linear mappings $f_i : V \rightarrow \mathbb{K}$ and therefore $f_i \in V^*$ for all i up to n . This of course generates a collection $\mathcal{B}^* := \{f_i\}_{i=1}^n \subseteq V^*$. We now claim that $\mathcal{B}^*$ is a basis for V^* .

We begin by showing that this is a linearly independent family. Assume now that $\sum_{i=1}^n \alpha_i f_i \equiv 0$; so in particular we have that

$$\sum_{i=1}^n \alpha_i f_i(b_j) = 0, \quad \forall j \in \{1, \dots, n\}$$

This gives us that $0 = \alpha_j f_j(b_j) = \alpha_j$ whence all the α_j are zero; thus $\mathcal{B}^*$ is a linearly independent family as was required. To complete the proof we shall now show that $\mathcal{B}^*$ is maximally independent. Indeed, assume that $f \in V^*$ is a linear functional on V into $\mathbb{K}$. Define now $\alpha_i = f(b_i)$ for all i . Now consider $\sum_{i=1}^n \alpha_i f_i$. Then this summation agrees with f on the basis $\mathcal{B}$ and by the argument in the previous proposition must be equal to f on V . Especially, f is a linear combination of the previous basis elements and we are done.

○

3.2 ROW AND COLUMNS SPACES

Some notation and language must first be introduced. Of course, as you will have likely seen in a high-school or Cégep algebra/linear-algebra course a system of *linear* equations is a collection of equations linear in m -variables:

$$\begin{cases} a_1x_1 + a_2x_2 + \cdots + a_mx_m = \alpha_1 \\ b_1x_1 + b_2x_2 + \cdots + b_mx_m = \alpha_2 \\ \vdots \\ n_1x_1 + n_2x_2 + \cdots + n_mx_m = \alpha_n \end{cases} \quad (17)$$

The goal of the “game” is to completely determine the solution space to the above system. That is, to characterize all m -tuples solving the system.

Our first observation is that the system in (17) may be written as a matrix. Indeed, the system above is equivalent to writing:

$$\begin{pmatrix} a_1 & \dots & a_m \\ \vdots & \ddots & \vdots \\ n_1 & \dots & n_m \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \quad (18)$$

Of course, the coefficient matrix $A = \begin{pmatrix} a_1 & \dots & a_m \\ \vdots & \ddots & \vdots \\ n_1 & \dots & n_m \end{pmatrix}$ is equivalent to a linear transformation T , and so we in fact may equivalently “solve” a linear transformation. In this section we illustrate several methods of solving these systems, from the point of view of matrix equations (mostly).

This is rather easy. Of course, we observe that if we have written this in the form $Ax = b$ where $A \in M_n(\mathbb{K})$ and $x, b \in \mathbb{K}^n$ are column vectors, it suffices to compute the inverse matrix. Indeed, if $\det(A) \neq 0$ then we know there exists an inverse matrix A^{-1} such that $I_n x = (A^{-1}A)x = A^{-1}(Ax) = A^{-1}b \in \mathbb{K}^n$. Moreover, since the inverse matrix is unique when it exists this is a unique solution.

There are several ways to compute this matrix A^{-1} provided it exists. Indeed, an “easy” method follows from ($\star_4$). Otherwise, one needs to reduce the matrix $A \in M_{n,m}(\mathbb{K})$ to the identity matrix $I_{n,m}$ via a series of elementary steps.

Definition 11. Let $A \in M_{m \times n}(\mathbb{K})$, the column space of A is the subspace spanned by the columns of A . Similarly, the row space is the subspace spanned by the rows of A .

Several observations are currently in order. Note first of all that the column space corresponds to the image of the linear transformation T induced by the matrix A . Now, it is natural to define the rank of a matrix. The row rank is the dimension of the row space while the column rank is defined to be the dimension of the column space. These are subsequently denoted $rk_R(A)$ and $rk_C(A)$ respectively.

COLOPHON

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